

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459962.7981 +/- 0.001 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.127 +/- 0.012
Transit depth (Rp/Rs)^2	1.6 +/- 0.3 [%]
Orbital Inclination (inc)	82.69 +/- 2.42
Airmass coefficient 1 (a1)	1.046 +/- 0.01
Airmass coefficient 2 (a2)	-0.038 +/- 0.016
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#1 - [407, 229]
Optimal Aperture	3.97
Optimal Annulus	6.6
Transit Duration (day)	0.1237 +/- 0.0057

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.127 ± 0.012 vs 0.1178 (deviation 0.77σ)
Duration fit vs published	0.1237 d vs 0.125 d (deviation 0.23σ)
Mid-time O-C	-2.6 min
Depth SNR	10.6
Residual scatter	0.99 %

Light curve

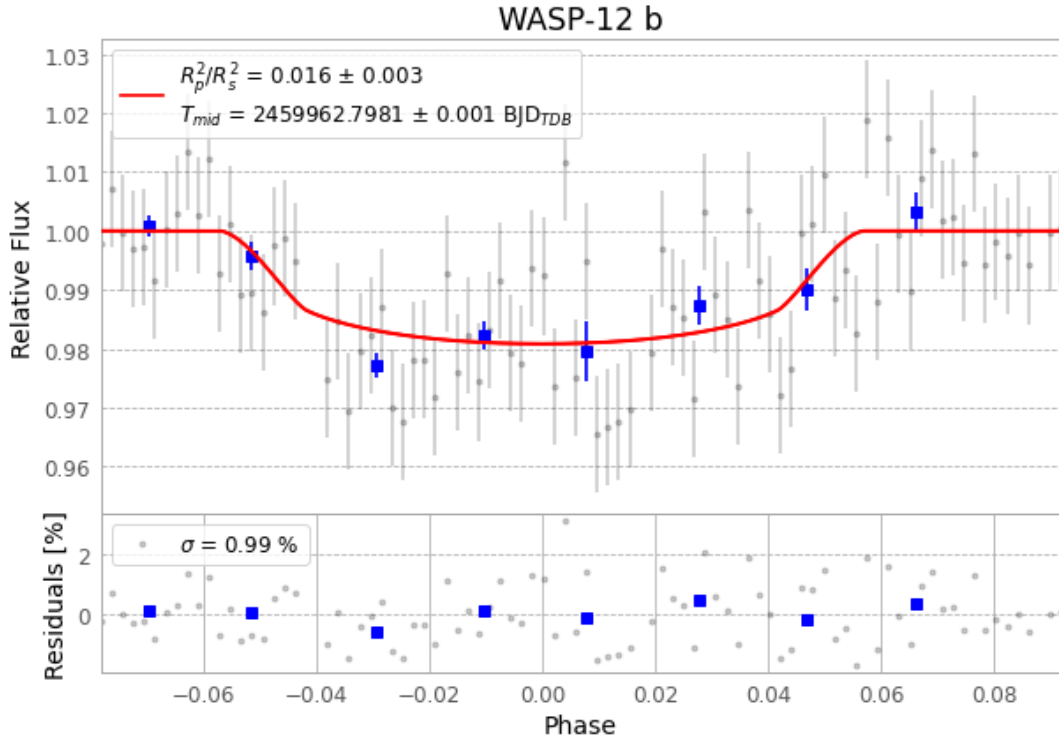


Figure 1 — FinalLightCurve_WASP-12 b_2023-01-17.png (SHA-256 893fa5629be1c948...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [425, 198], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8fb0a85f12bb4d6a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

95 FITS frames (62 MB), WASP-12230118045715.FITS → WASP-12230118093915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230118.log (b374b9ad38d9...), FinalLightCurve_WASP-12 b_2023-01-17.png (893fa5629be1...), FinalLightCurve_WASP-12 b_2023-01-17.csv (22caa8720c98...)

Compute resources

Wall time 380.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 03:05Z.